

Interleaflet coupling drives the thermotropic behaviour of asymmetric membranes

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ABSTRACT

Interleaflet coupling is central to membrane organization and function. Registration of ordered domains has been proposed to generate transbilayer signalling platforms, while ordered lipid environments can regulate membrane-protein function through selective recruitment. Yet, how such coupling operates in asymmetric membranes remains incompletely understood. Experiments have shown that, in asymmetric bilayers, liquid-ordered domain formation can be induced or suppressed by the composition of the apposing leaflet, demonstrating composition-dependent coupling¹. More recently, asymmetric POPC/POPE vesicles revealed a distinct cooling-induced coupling, in which leaflet melting transitions are either cooperative or decoupled depending on lipid sidedness and vesicle curvature².

Here, we use molecular dynamics simulations to study the thermotropic behavior of POPC and POPE membranes with controlled compositional and numerical asymmetry. We developed a simulated-cooling protocol based on the sticky-tape setup³, allowing asymmetric membranes to relax both area strain and bending energy during the liquid-to-gel transition. In compositionally symmetric membranes, numerical asymmetry induces curvature but produces only weak leaflet decoupling. In POPC^{up}/POPE^{down} membranes, instead, numerical asymmetry controls the mismatch between leaflet order parameters and determines whether melting is coupled or decoupled. The strongest decoupling occurs when the POPC leaflet is lipid-depleted, in agreement with the curvature-dependent behavior observed experimentally². This decoupling is enabled by the stabilization of an intermediate POPC phase with liquid-like dynamics, partially ordered tails, and strongly reduced differential stress.

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